Fase 1: Preparación del entorno local con Docker

Evidencia de ejecución

Se levantó correctamente el ecosistema local utilizando Docker Compose.

Servicios levantados:

- Kafka Broker
- Kafka UI
- Frontend de pruebas

Se validó el acceso a Kafka UI mediante:

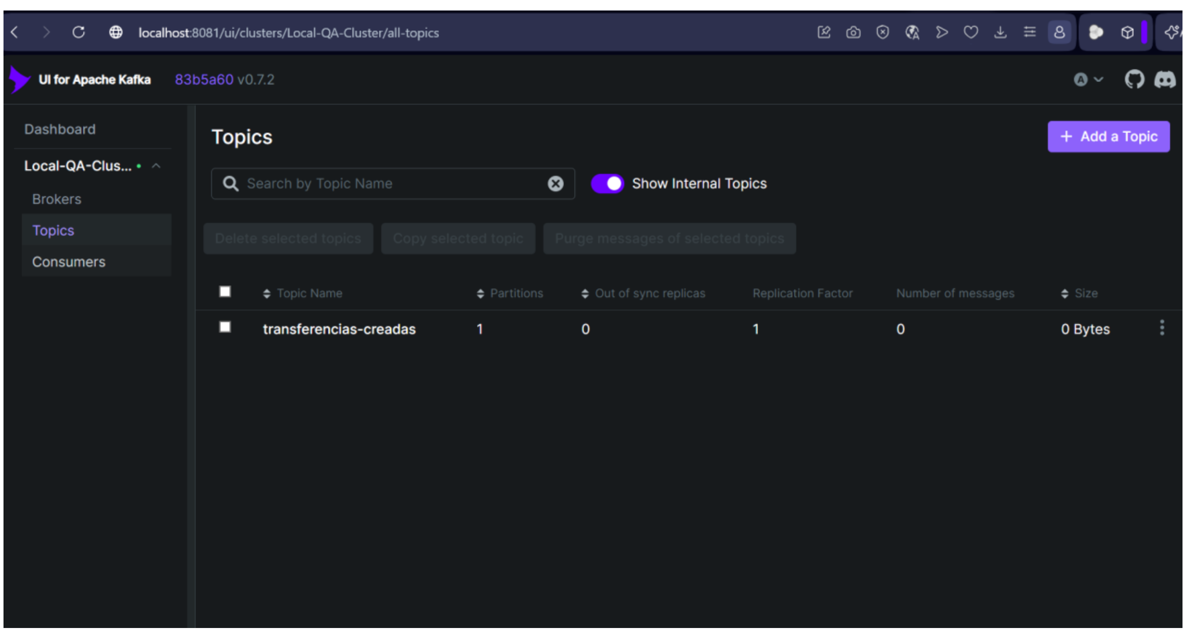
http://localhost:8081

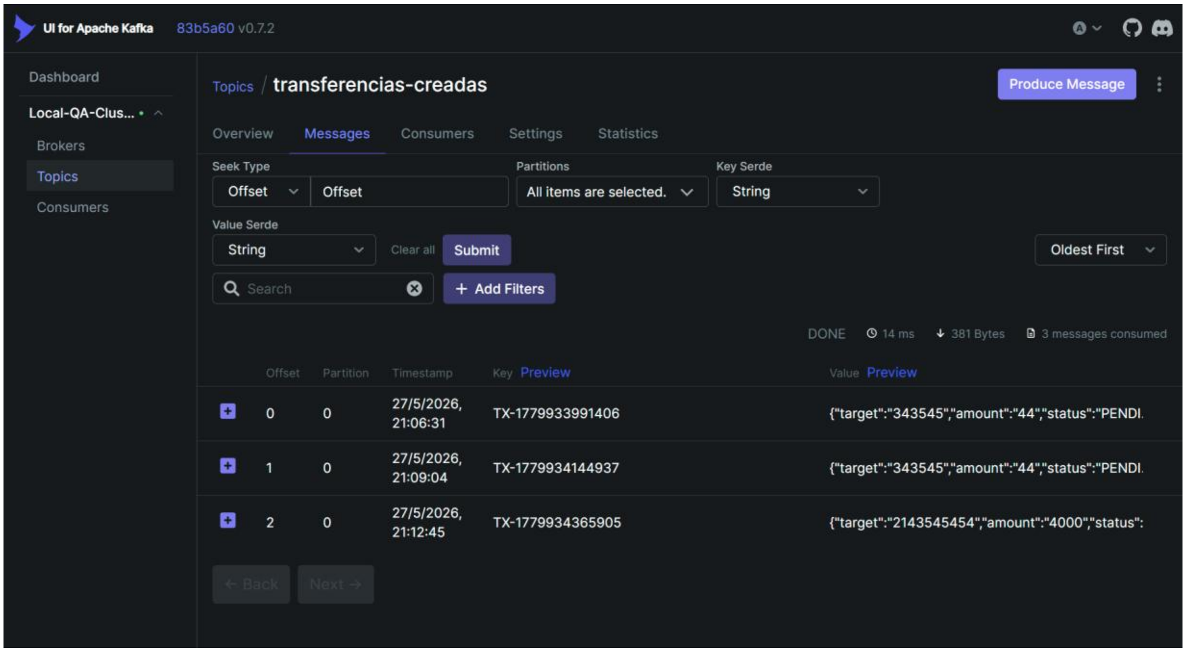
Posteriormente se creó manualmente el tópico:

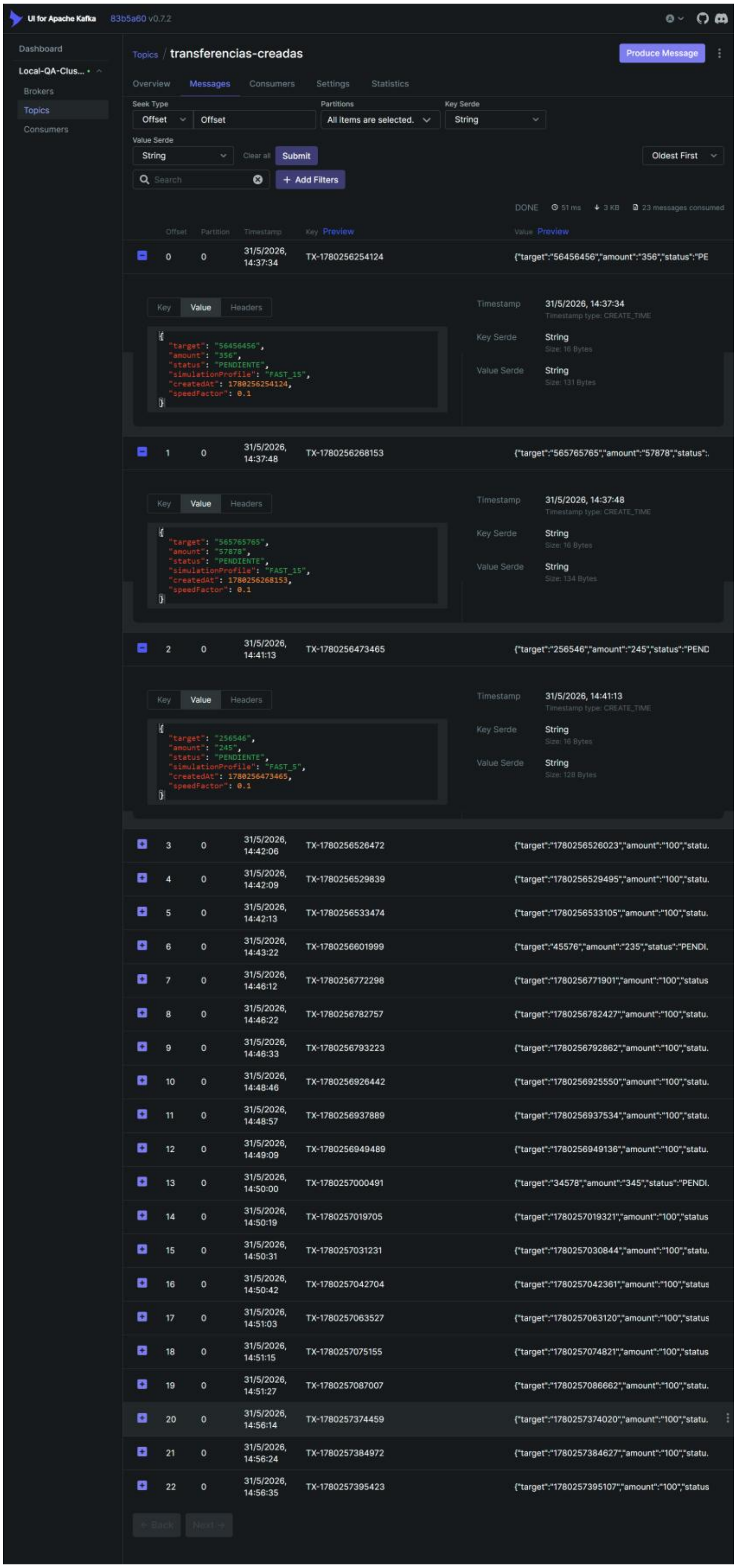
transferencias-creadas

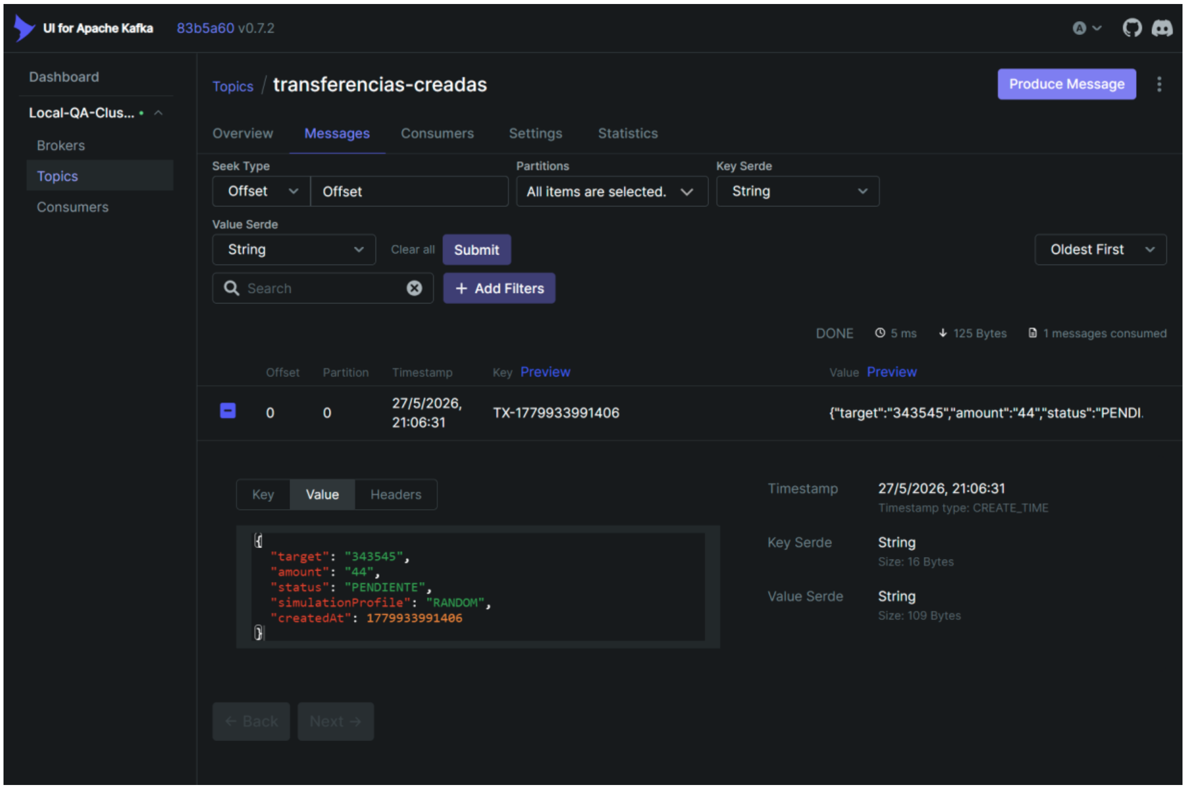
como parte de la preparación del flujo asíncrono basado en Kafka.

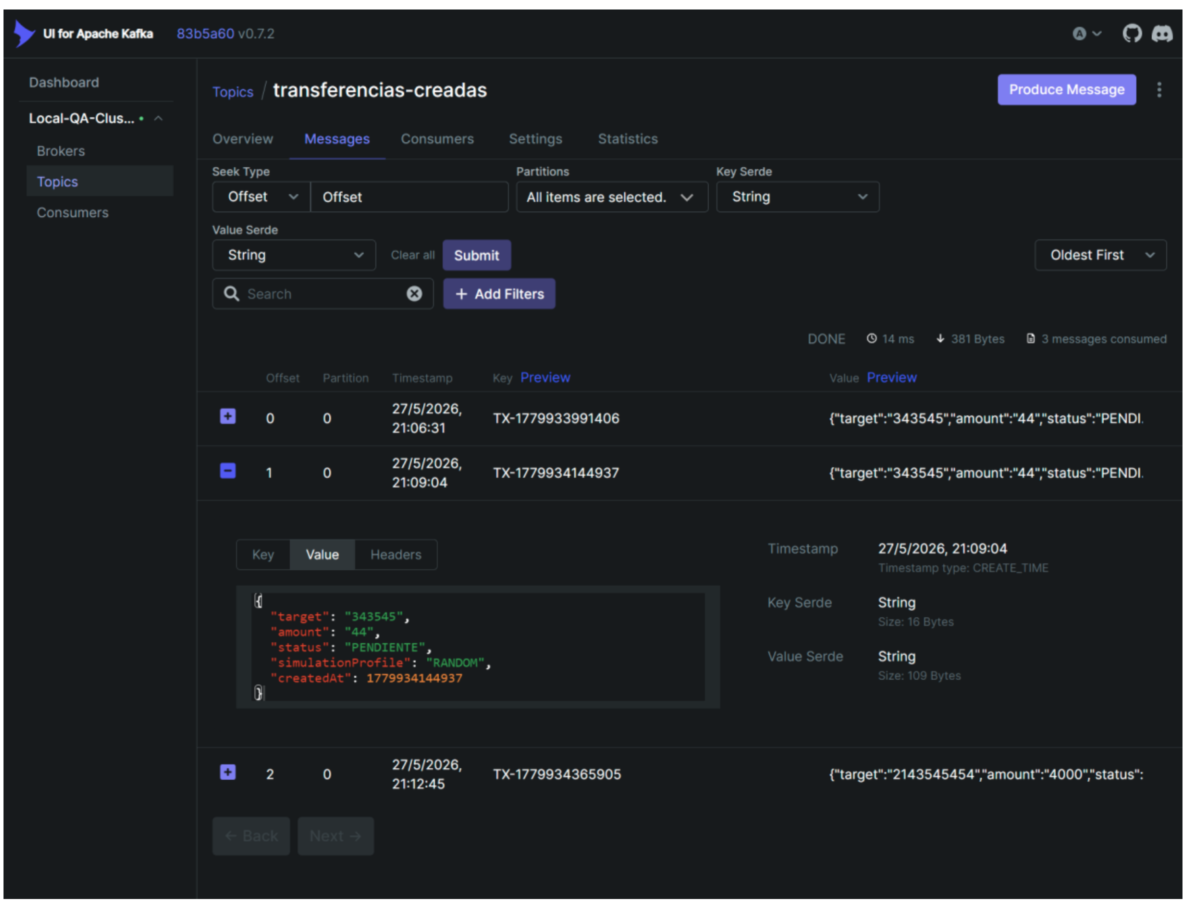
Evidencia




UI for Apache Kafka
83b5a60 v0.7.2
Dashboard
Local-QA-Clus...
Brokers
Topics
Consumers
Topics / transferencias-creadas
Produce Message
Overview
Messages
Consumers
Settings
Statistics
Seek Type
Offset
Offset
Partitions
All items are selected.
Key Serde
String
Value Serde
String
Clear all
Submit
Oldest First
Search
+ Add Filters
DONE
14 ms
381 Bytes
3 messages consumed
Offset
Partition
Timestamp
Key Preview
Value Preview
0
0
27/5/2026, 21:06:31
TX-1779933991406
{"target":"343545","amount":"44","status":"PENDI.
1
0
27/5/2026, 21:09:04
TX-1779934144937
{"target":"343545","amount":"44","status":"PENDI.
2
0
27/5/2026, 21:12:45
TX-1779934365905
{"target":"2143545454","amount":"4000","status":
← Back
Next →


UI for Apache Kafka
83b5a60 v0.7.2
Dashboard
Local-QA-Clus...
Brokers
Topics
Consumers
Topics / transferencias-creadas
Produce Message
Overview
Messages
Consumers
Settings
Statistics
Seek Type
Offset
Offset
Partitions
All items are selected.
Key Serde
String
Value Serde
String
Clear all
Submit
Oldest First
Search
+ Add Filters
DONE
51 ms
3 KB
23 messages consumed
Offset
Partition
Timestamp
Key Preview
Value Preview
0 0 31/5/2026, 14:37:34 TX-1780256254124 {"target":"56456456","amount":"356","status":"PE
Key
Value
Headers
{
"target": "56456456",
"amount": "356",
"status": "PENDIENTE",
"simulationProfile": "FAST_15",
"createdAt": 1780256254124,
"speedFactor": 0.1
}
Timestamp
31/5/2026, 14:37:34
Timestamp type: CREATE_TIME
Key Serde
String
Size: 16 Bytes
Value Serde
String
Size: 131 Bytes
1 0 31/5/2026, 14:37:48 TX-1780256268153 {"target":"565765765","amount":"57878","status":.
Key
Value
Headers
{
"target": "565765765",
"amount": "57878",
"status": "PENDIENTE",
"simulationProfile": "FAST_15",
"createdAt": 1780256268153,
"speedFactor": 0.1
}
Timestamp
31/5/2026, 14:37:48
Timestamp type: CREATE_TIME
Key Serde
String
Size: 16 Bytes
Value Serde
String
Size: 134 Bytes
2 0 31/5/2026, 14:41:13 TX-1780256473465 {"target":"256546","amount":"245","status":"PEND
Key
Value
Headers
{
"target": "256546",
"amount": "245",
"status": "PENDIENTE",
"simulationProfile": "FAST_5",
"createdAt": 1780256473465,
"speedFactor": 0.1
}
Timestamp
31/5/2026, 14:41:13
Timestamp type: CREATE_TIME
Key Serde
String
Size: 16 Bytes
Value Serde
String
Size: 128 Bytes
3 0 31/5/2026, 14:42:06 TX-1780256526472 {"target":"1780256526023","amount":"100","statu.
4 0 31/5/2026, 14:42:09 TX-1780256529839 {"target":"1780256529495","amount":"100","statu.
5 0 31/5/2026, 14:42:13 TX-1780256533474 {"target":"1780256533105","amount":"100","statu.
6 0 31/5/2026, 14:43:22 TX-1780256601999 {"target":"45576","amount":"235","status":"PENDI.
7 0 31/5/2026, 14:46:12 TX-1780256772298 {"target":"1780256771901","amount":"100","status
8 0 31/5/2026, 14:46:22 TX-1780256782757 {"target":"1780256782427","amount":"100","statu.
9 0 31/5/2026, 14:46:33 TX-1780256793223 {"target":"1780256792862","amount":"100","statu.
10 0 31/5/2026, 14:48:46 TX-1780256926442 {"target":"1780256925550","amount":"100","statu.
11 0 31/5/2026, 14:48:57 TX-1780256937889 {"target":"1780256937534","amount":"100","statu.
12 0 31/5/2026, 14:49:09 TX-1780256949489 {"target":"1780256949136","amount":"100","statu.
13 0 31/5/2026, 14:50:00 TX-1780257000491 {"target":"34578","amount":"345","status":"PENDI.
14 0 31/5/2026, 14:50:19 TX-1780257019705 {"target":"1780257019321","amount":"100","status
15 0 31/5/2026, 14:50:31 TX-1780257031231 {"target":"1780257030844","amount":"100","statu.
16 0 31/5/2026, 14:50:42 TX-1780257042704 {"target":"1780257042361","amount":"100","status
17 0 31/5/2026, 14:51:03 TX-1780257063527 {"target":"1780257063120","amount":"100","status
18 0 31/5/2026, 14:51:15 TX-1780257075155 {"target":"1780257074821","amount":"100","status
19 0 31/5/2026, 14:51:27 TX-1780257087007 {"target":"1780257086662","amount":"100","statu.
20 0 31/5/2026, 14:56:14 TX-1780257374459 {"target":"1780257374020","amount":"100","statu.
21 0 31/5/2026, 14:56:24 TX-1780257384972 {"target":"1780257384627","amount":"100","statu.
22 0 31/5/2026, 14:56:35 TX-1780257395423 {"target":"1780257395107","amount":"100","status
Back
Next


UI for Apache Kafka
83b5a60 v0.7.2
Dashboard
Local-QA-Clus...
Brokers
Topics
Consumers
Topics / transferencias-creadas
Produce Message
Overview
Messages
Consumers
Settings
Statistics
Seek Type
Offset
Offset
Partitions
All items are selected.
Key Serde
String
Value Serde
String
Clear all
Submit
Oldest First
Search
+ Add Filters
DONE
5 ms
125 Bytes
1 messages consumed
Offset
Partition
Timestamp
Key Preview
Value Preview
0
0
27/5/2026, 21:06:31
TX-1779933991406
{"target":"343545","amount":"44","status":"PENDI
Key
Value
Headers
{
"target": "343545",
"amount": "44",
"status": "PENDIENTE",
"simulationProfile": "RANDOM",
"createdAt": 1779933991406
}
Timestamp
27/5/2026, 21:06:31
Timestamp type: CREATE_TIME
Key Serde
String
Size: 16 Bytes
Value Serde
String
Size: 109 Bytes
← Back
Next →


UI for Apache Kafka
83b5a60 v0.7.2
Dashboard
Local-QA-Clus...
Brokers
Topics
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Topics / transferencias-creadas
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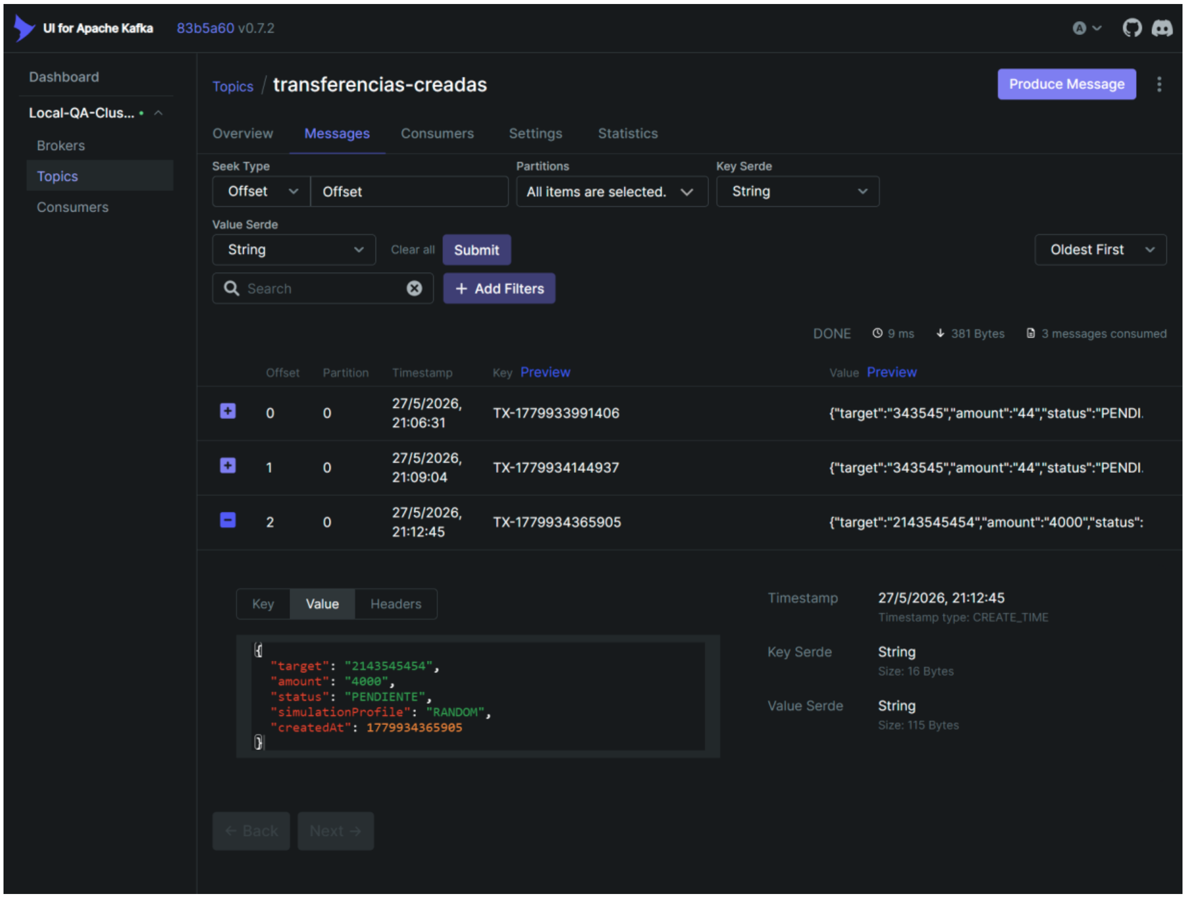
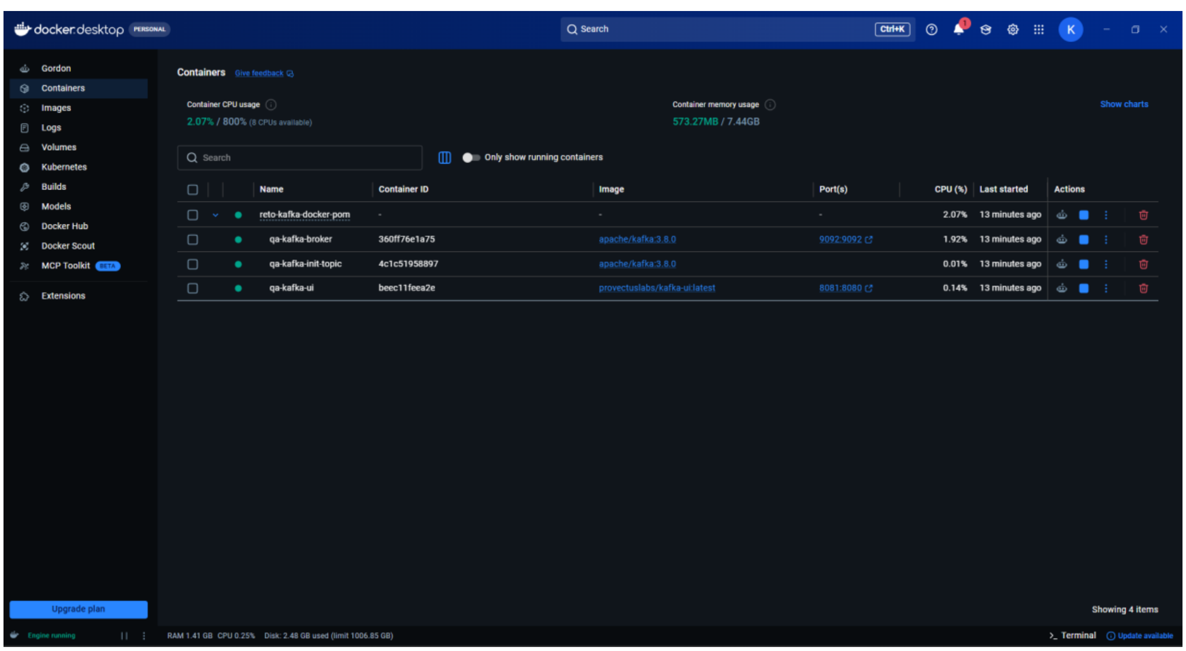
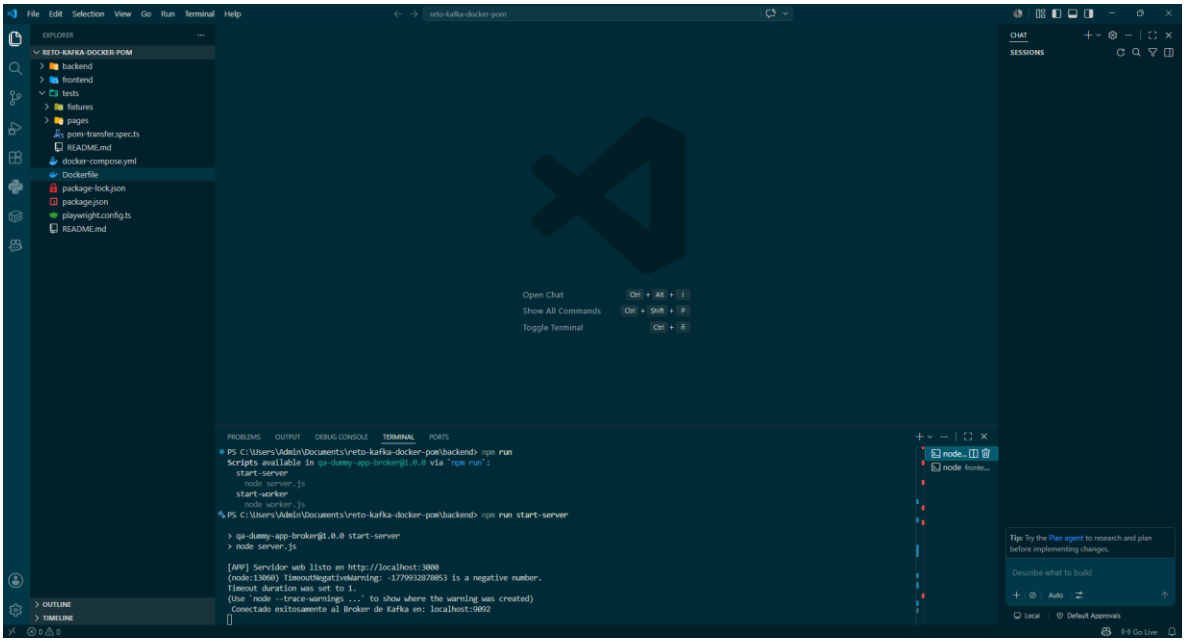
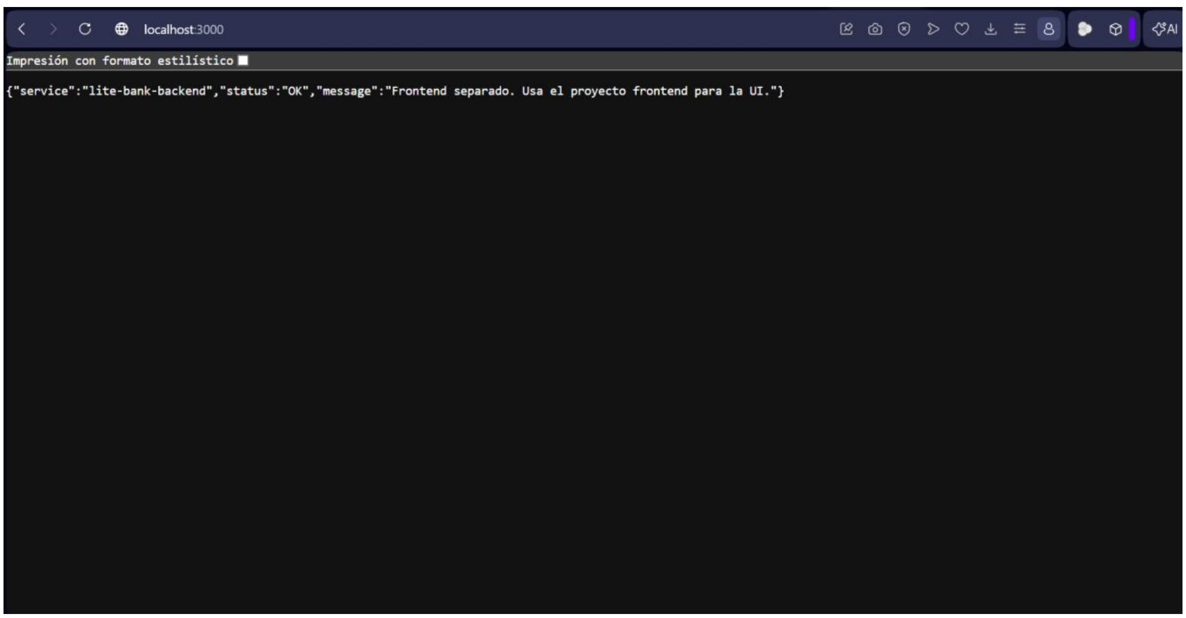
All items are selected.
Key Serde
String
Value Serde
String
Clear all
Submit
Oldest First
Search
+ Add Filters
DONE
14 ms
381 Bytes
3 messages consumed
Offset
Partition
Timestamp
Key Preview
Value Preview
0
0
27/5/2026, 21:06:31
TX-1779933991406
{"target":"343545","amount":"44","status":"PENDI
1
0
27/5/2026, 21:09:04
TX-1779934144937
{"target":"343545","amount":"44","status":"PENDI
Key
Value
Headers
{
"target": "343545",
"amount": "44",
"status": "PENDIENTE",
"simulationProfile": "RANDOM",
"createdAt": 1779934144937
}
Timestamp
27/5/2026, 21:09:04
Timestamp type: CREATE_TIME
Key Serde
String
Size: 16 Bytes
Value Serde
String
Size: 109 Bytes
2
0
27/5/2026, 21:12:45
TX-1779934365905
{"target":"2143545454","amount":"4000","status":
← Back
Next →


UI for Apache Kafka
83b5a60 v0.7.2
Dashboard
Local-QA-Clus...
Brokers
Topics
Consumers
Topics / transferencias-creadas
Produce Message
Overview
Messages
Consumers
Settings
Statistics
Seek Type
Offset
Offset
Partitions
All items are selected.
Key Serde
String
Value Serde
String
Clear all
Submit
Oldest First
Search
Add Filters
DONE
9 ms
381 Bytes
3 messages consumed
Offset
Partition
Timestamp
Key Preview
Value Preview
0 0 27/5/2026, 21:06:31 TX-1779933991406 {"target":"343545","amount":"44","status":"PENDI
1 0 27/5/2026, 21:09:04 TX-1779934144937 {"target":"343545","amount":"44","status":"PENDI
2 0 27/5/2026, 21:12:45 TX-1779934365905 {"target":"2143545454","amount":"4000","status":
Key
Value
Headers
{
"target": "2143545454",
"amount": "4000",
"status": "PENDIENTE",
"simulationProfile": "RANDOM",
"createdAt": 1779934365905
}
Timestamp
27/5/2026, 21:12:45
Timestamp type: CREATE_TIME
Key Serde
String
Size: 16 Bytes
Value Serde
String
Size: 115 Bytes
Back
Next

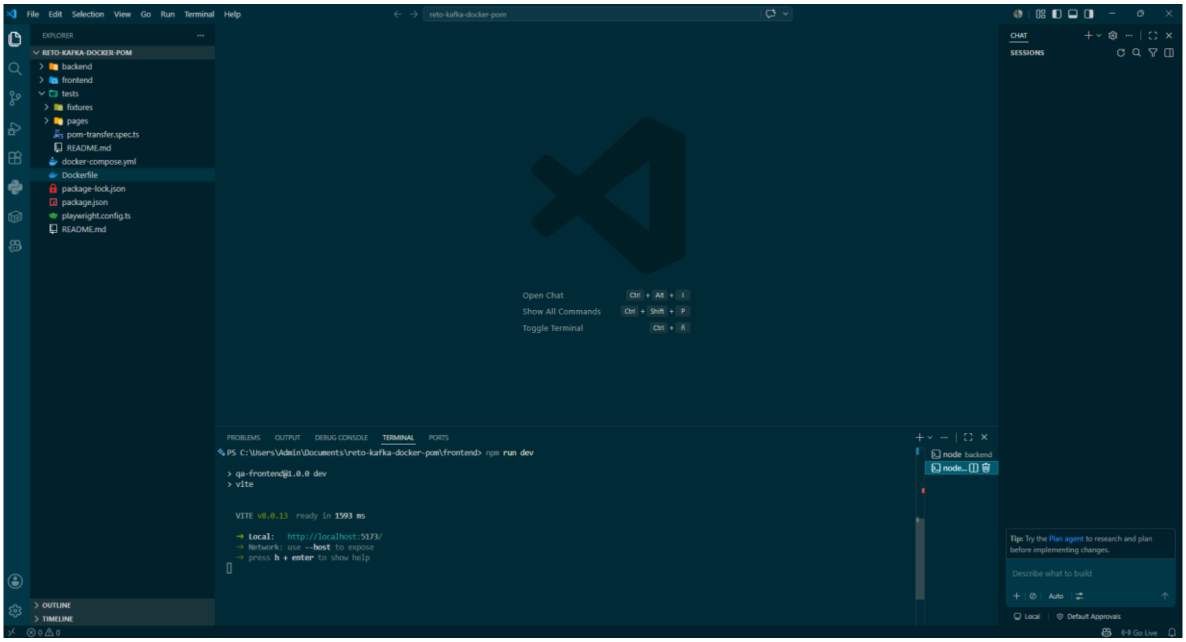

docker desktop PERSONAL
Search
Ctrl+K
Gordon
Containers
Images
Logs
Volumes
Kubernetes
Builds
Models
Docker Hub
Docker Scout
MCP Toolkit BETA
Extensions
Containers Give feedback
Container CPU usage
2.07% / 800% (8 CPUs available)
Container memory usage
573.27MB / 7.44GB
Show charts
Search
Only show running containers
Name
Container ID
Image
Port(s)
CPU (%)
Last started
Actions
reto-kafka-docker-pom - - - 2.07% 13 minutes ago
qa-kafka-broker 360ff76e1a75 apache/kafka:3.8.0 9092:9092 1.92% 13 minutes ago
qa-kafka-init-topic 4c1c51958897 apache/kafka:3.8.0 0.01% 13 minutes ago
qa-kafka-ui beec11feea2e provectuslabs/kafka-ui:latest 8081:8080 0.14% 13 minutes ago
Showing 4 items
Upgrade plan
Engine running
RAM 1.41 GB CPU 0.25% Disk: 2.48 GB used (limit 1006.85 GB)
Terminal
Update available

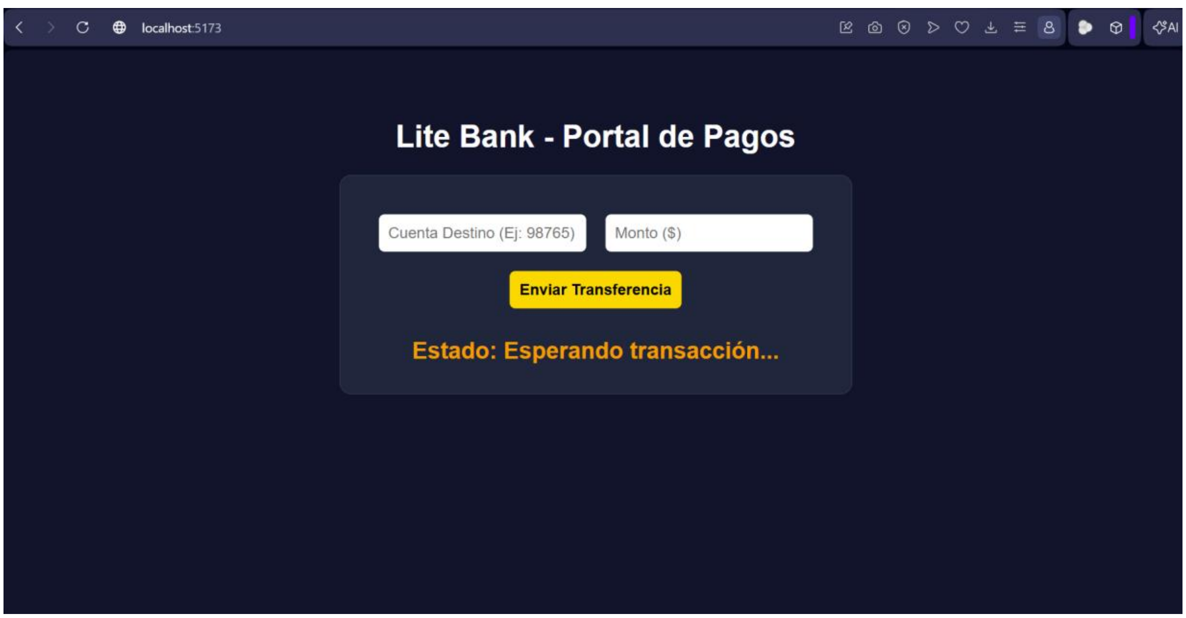


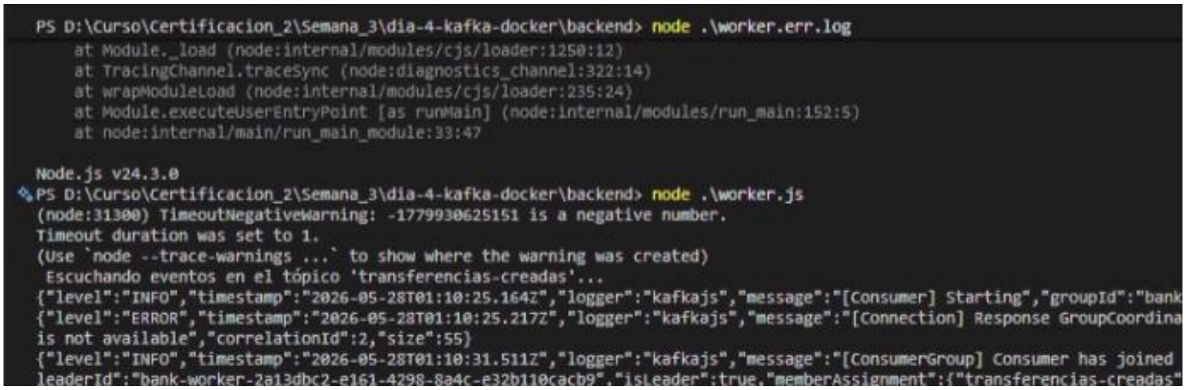
File Edit Selection View Go Run Terminal Help
reto-kafka-docker-pom
EXPLORER
RETO-KAFKA-DOCKER-POM
backend
frontend
tests
fixtures
pages
pom-transfer.spec.ts
README.md
docker-compose.yml
Dockerfile
package-lock.json
package.json
playwright.config.ts
README.md
OUTLINE
TIMELINE
Open Chat
Show All Commands
Toggle Terminal
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Admin\Documents\reto-kafka-docker-pom\backend> npm run
Scripts available in qa-dummy-app-broker@1.0.0 via `npm run`:
start-server
node server.js
start-worker
node worker.js
PS C:\Users\Admin\Documents\reto-kafka-docker-pom\backend> npm run start-server
> qa-dummy-app-broker@1.0.0 start-server
> node server.js
[APP] Servidor web listo en http://localhost:3000
(node:13060) TimeoutNegativeWarning: -1779932878053 is a negative number.
Timeout duration was set to 1.
(Use `node --trace-warnings ...` to show where the warning was created)
Conectado exitosamente al Broker de Kafka en: localhost:9092
CHAT
SESSIONS
Tip: Try the Plan agent to research and plan before implementing changes.
Describe what to build
Auto
Local
Default Approvals
Go Live

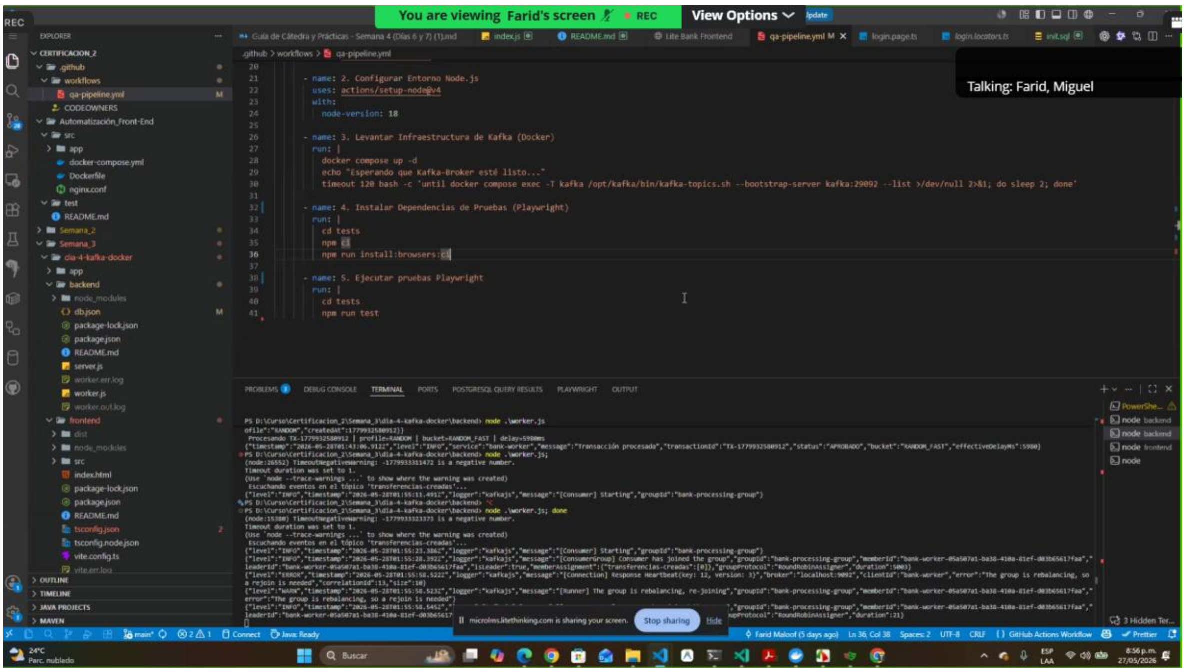


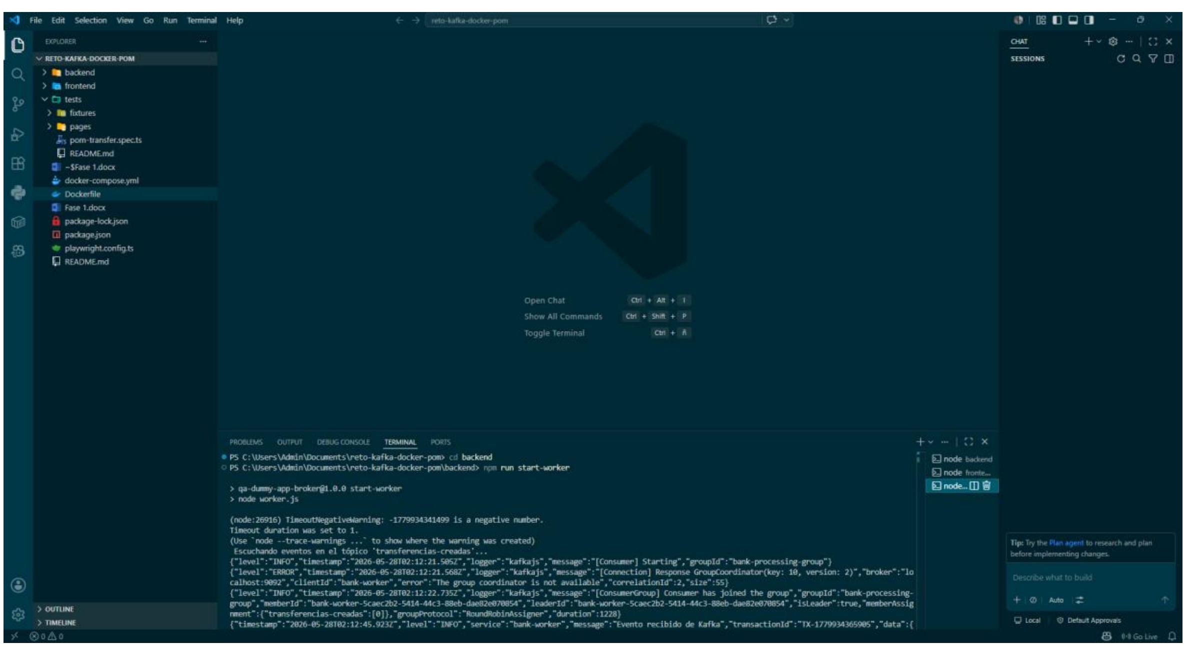
localhost:3000
Impresión con formato estilístico
{"service":"lite-bank-backend","status":"OK","message":"Frontend separado. Usa el proyecto frontend para la UI."}


File Edit Selection View Go Run Terminal Help
reto-kafka-docker-pom
EXPLORER
RETO-KAFKA-DOCKER-POM
backend
frontend
tests
fixtures
pages
pom-transfer.spec.ts
README.md
docker-compose.yml
Dockerfile
package-lock.json
package.json
playwright.config.ts
README.md
Open Chat
Show All Commands
Toggle Terminal
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Admin\Documents\reto-kafka-docker-pom\frontend> npm run dev
> vite
Local: http://localhost:5173/
CHAT
SESSIONS
Describe what to build
OUTLINE
TIMELINE


localhost:5173
Lite Bank - Portal de Pagos
Cuenta Destino (Ej: 98765)
Monto ($)
Enviar Transferencia
Estado: Esperando transacción...


PS D:\Curso\Certificacion_2\Semana_3\dia-4-kafka-docker\backend> node .\worker.err.log
at Module._load (node:internal/modules/cjs/loader:1250:12)
at TracingChannel.traceSync (node:diagnostics_channel:322:14)
at wrapModuleLoad (node:internal/modules/cjs/loader:235:24)
at Module.executeUserEntryPoint [as runMain] (node:internal/modules/run_main:152:5)
at node:internal/main/run_main_module:33:47
Node.js v24.3.0
PS D:\Curso\Certificacion_2\Semana_3\dia-4-kafka-docker\backend> node .\worker.js
(node:31300) TimeoutNegativeWarning: -1779930625151 is a negative number.
Timeout duration was set to 1.
(Use `node --trace-warnings ...` to show where the warning was created)
Escuchando eventos en el tópico 'transferencias-creadas'...
{"level":"INFO","timestamp":"2026-05-28T01:10:25.164Z","logger":"kafkajs","message":"[Consumer] Starting","groupId":"bank
{"level":"ERROR","timestamp":"2026-05-28T01:10:25.217Z","logger":"kafkajs","message":"[Connection] Response GroupCoordina
is not available","correlationId":2,"size":55}
{"level":"INFO","timestamp":"2026-05-28T01:10:31.511Z","logger":"kafkajs","message":"[ConsumerGroup] Consumer has joined
leaderId":"bank-worker-2a13dbc2-e161-4298-8a4c-e32b110cacb9","isLeader":true,"memberAssignment":{"transferencias-creadas"


You are viewing Farid's screen
REC
View Options
Talking: Farid, Miguel
qa-pipeline.yml
- name: 2. Configurar Entorno Node.js
uses: actions/setup-node@v4
with:
node-version: 18
- name: 3. Levantar Infraestructura de Kafka (Docker)
run: |
docker compose up -d
echo "Esperando que Kafka-Broker esté listo..."
timeout 120 bash -c 'until docker compose exec -T kafka /opt/kafka/bin/kafka-topics.sh --bootstrap-server kafka:29092 --list >/dev/null 2>&1; do sleep 2; done'
- name: 4. Instalar Dependencias de Pruebas (Playwright)
run: |
cd tests
npm ci
npm run install:browsers:ci
- name: 5. Ejecutar pruebas Playwright
run: |
cd tests
npm run test
Stop sharing


RETO-KAFKA-DOCKER-POM
Open Chat
Show All Commands
Toggle Terminal
PS C:\Users\Admin\Documents\reto-kafka-docker-pom> cd backend
PS C:\Users\Admin\Documents\reto-kafka-docker-pom\backend> npm run start-worker
> qa-dummy-app-broker@1.0.0 start-worker
> node worker.js
(node:26916) TimeoutNegativeWarning: -1779934341499 is a negative number.
Timeout duration was set to 1.
(Use `node --trace-warnings ...` to show where the warning was created)
Escuchando eventos en el tópico 'transferencias-creadas'...
{"level":"INFO","timestamp":"2026-05-28T02:12:21.505Z","logger":"kafkajs","message":"[Consumer] Starting","groupId":"bank-processing-group"}
{"level":"ERROR","timestamp":"2026-05-28T02:12:21.568Z","logger":"kafkajs","message":"[Connection] Response GroupCoordinator(key: 10, version: 2)","broker":"localhost:9092","clientId":"bank-worker","error":"The group coordinator is not available","correlationId":2,"size":55}
{"level":"INFO","timestamp":"2026-05-28T02:12:22.735Z","logger":"kafkajs","message":"[ConsumerGroup] Consumer has joined the group","groupId":"bank-processing-group","memberId":"bank-worker-5caec2b2-5414-44c3-88eb-dae82e070854","leaderId":"bank-worker-5caec2b2-5414-44c3-88eb-dae82e070854","isLeader":true,"memberAssignment":{"transferencias-creadas":[0]},"groupProtocol":"RoundRobinAssigner","duration":1228}
{"timestamp":"2026-05-28T02:12:45.923Z","level":"INFO","service":"bank-worker","message":"Evento recibido de Kafka","transactionId":"TX-1779934365905","data":{

Fase 2: Automatización resiliente con Playwright

Evidencia de implementación

Se implementó una prueba automatizada utilizando el framework de automatización **Playwright con JavaScript**, aplicando el patrón **Page Object Model (POM)** sobre el flujo asíncrono de transferencias de la aplicación.

Archivos implementados

```
tests/
    pages/
        TransferPage.js
    transfer-async.spec.js
```

Descripción del Patrón POM

Se creó la clase TransferPage, donde se centralizaron los localizadores y acciones principales:

- Campo Cuenta Destino
- Campo Monto
- Botón Enviar Transferencia
- Cuadro de estado #status-box

Lógica del test creado

El test realiza el siguiente flujo:

1. Navega a la aplicación.
2. Ingresa datos únicos de prueba.
3. Envía la transferencia.
4. Espera dinámicamente hasta que se valida el estado cambie de PENDIENTE a APROBADO

Polling inteligente

Se implementó polling dinámico con Playwright utilizando expect.poll, sin usar esperas duras como sleep() o waitForTimeout().

Configuración aplicada:

timeout: 10000,

intervals: [500, 1000]

Esto permite consultar dinámicamente el estado hasta que el cuadro muestre APROBADO, con un tiempo máximo de 10 segundos.

Ejecución consecutiva

Se utilizo el siguiente comando para realizar la ejecución 3 veces consecutivas sin fallos

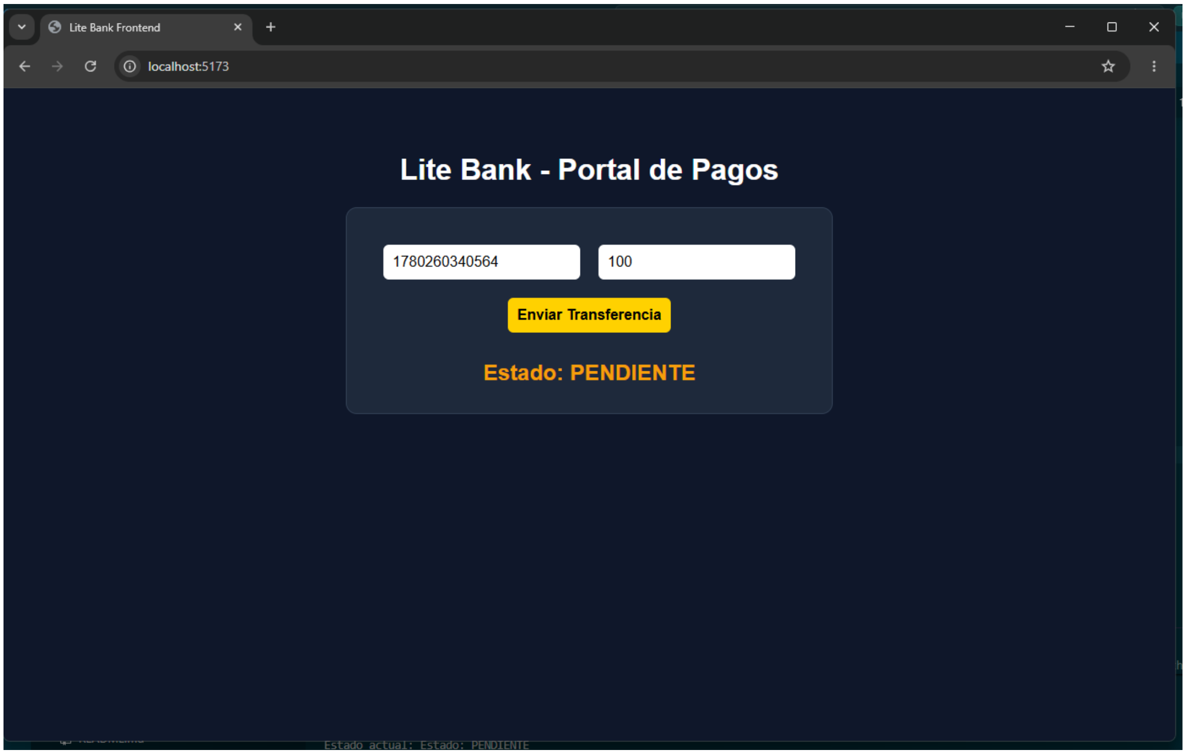
npx playwright test tests/transfer-async.spec.js --headed --workers=1 --repeat-each=3

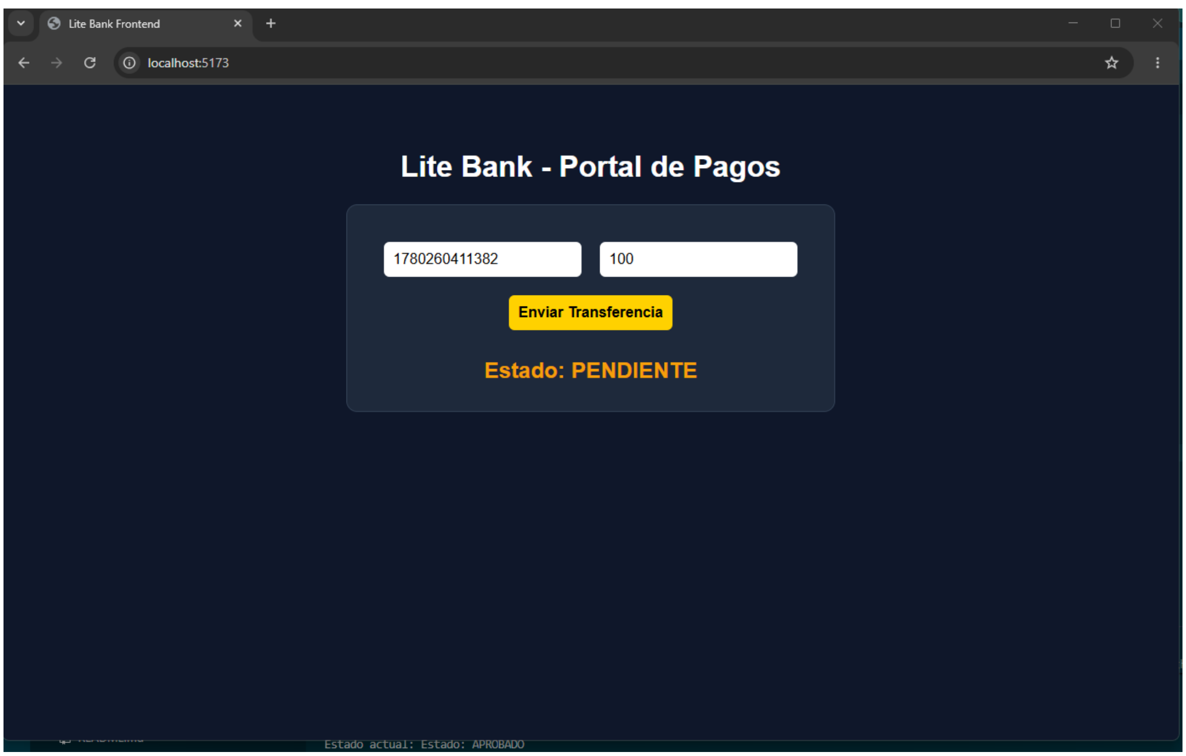
Observaciones

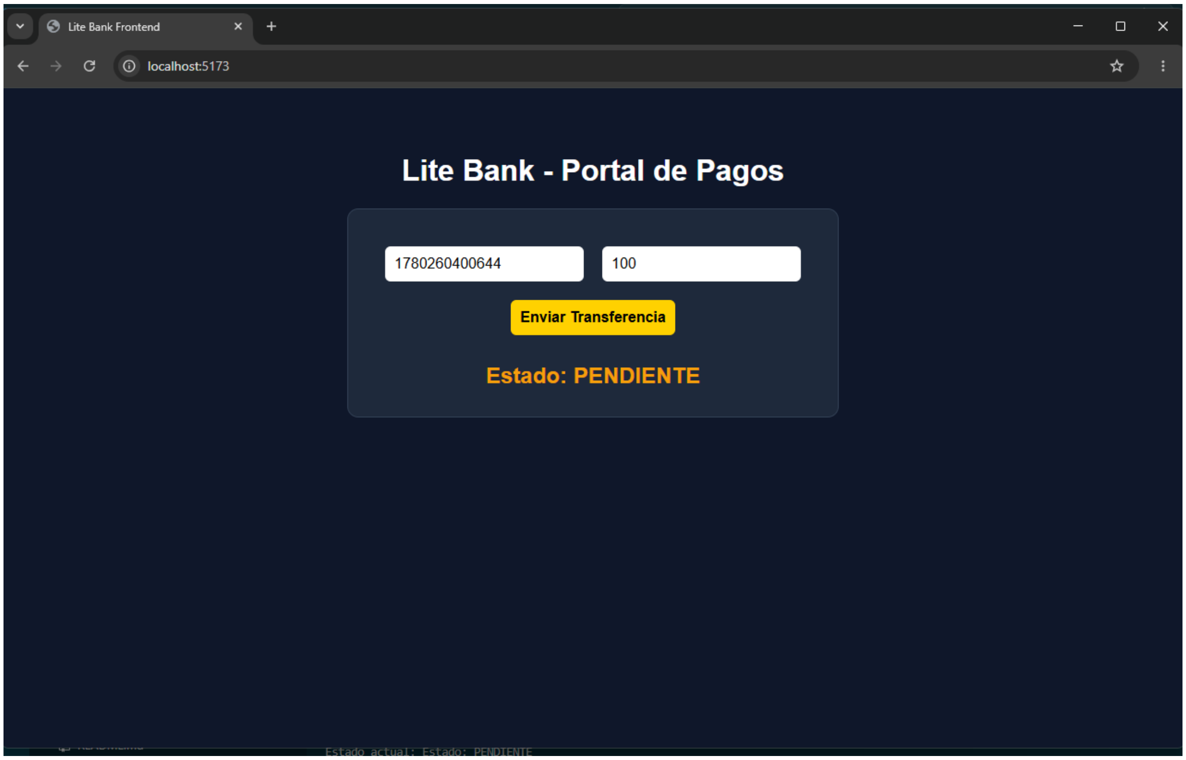
Durante las pruebas iniciales se identificó cambios en los tiempos de procesamiento, se identifica que posiblemente sea al uso del perfil de simulación RANDOM, el cual generaba demoras superiores al timeout definido para la prueba.

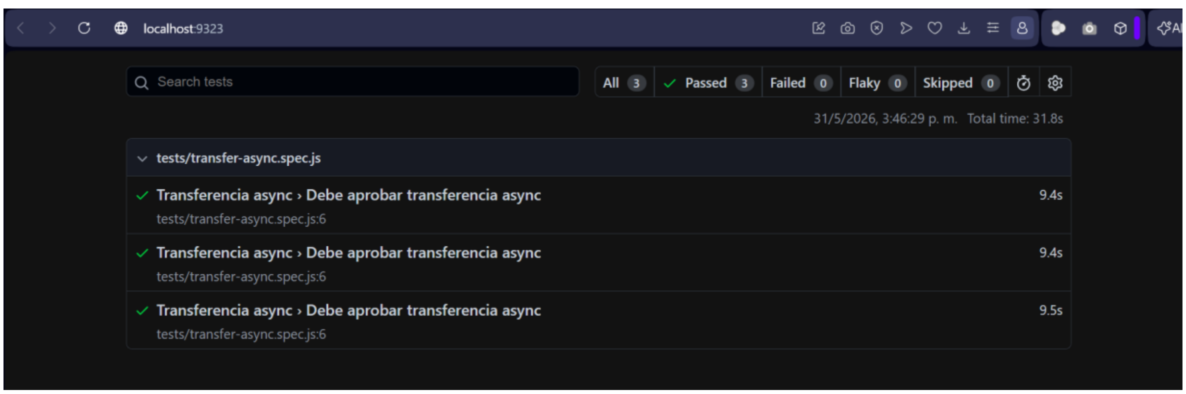
Para garantizar la estabilidad de la automatización y evitar comportamientos intermitentes (flakiness), se configuró un perfil de procesamiento controlado (FAST_15 con speedFactor ajustado), permitiendo que el estado final de la transacción se alcanzara consistentemente dentro del tiempo máximo establecido.

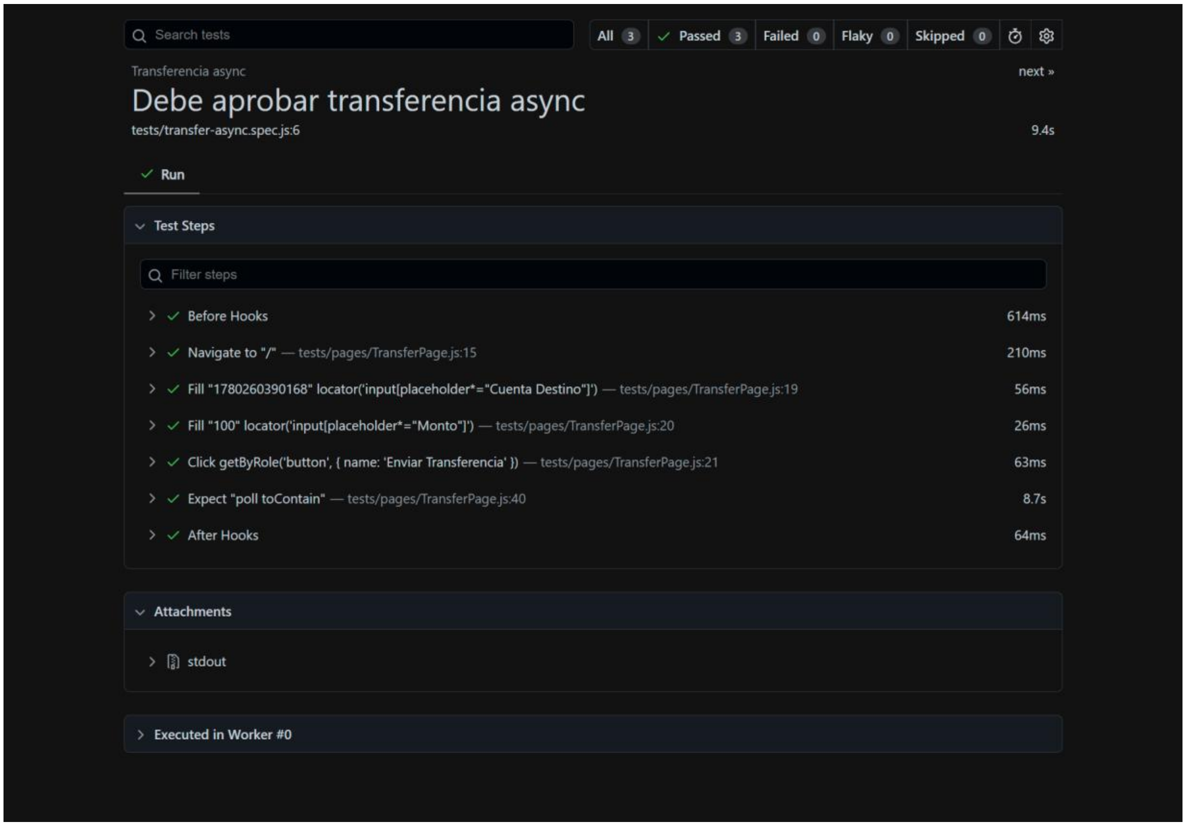
La estrategia de validación implementada mediante polling inteligente (expect.poll) permitió que la prueba esperara dinámicamente el cambio de estado sin utilizar esperas fijas (sleep), reduciendo el riesgo de falsos positivos y falsos negativos en la ejecución.


Lite Bank Frontend
localhost:5173
Lite Bank - Portal de Pagos
1780260340564
100
Enviar Transferencia
Estado: PENDIENTE
Estado actual: Estado: PENDIENTE


Lite Bank Frontend
localhost:5173
Lite Bank - Portal de Pagos
1780260411382
100
Enviar Transferencia
Estado: PENDIENTE
Estado actual: Estado: APROBADO


Lite Bank Frontend
localhost:5173
Lite Bank - Portal de Pagos
1780260400644
100
Enviar Transferencia
Estado: PENDIENTE
Estado actual: Estado: PENDIENTE


localhost:9323
Search tests
All 3
Passed 3
Failed 0
Flaky 0
Skipped 0
31/5/2026, 3:46:29 p. m. Total time: 31.8s
tests/transfer-async.spec.js
Transferencia async › Debe aprobar transferencia async 9.4s
tests/transfer-async.spec.js:6
Transferencia async › Debe aprobar transferencia async 9.4s
tests/transfer-async.spec.js:6
Transferencia async › Debe aprobar transferencia async 9.5s
tests/transfer-async.spec.js:6



Fase 3: Orquestación en la nube (CI/CD – GITHUB ACTIONS)

Objetivo

Automatizar la ejecución de las pruebas Playwright mediante un pipeline CI/CD ejecutado desde GitHub Actions, garantizando la validación automática del flujo de transferencia asíncrona ante cada cambio realizado en el repositorio.

Actividades realizadas

1. Creación del repositorio GitHub

Se creó el repositorio remoto para alojar el proyecto de automatización:

Repositorio:

[reto-uno-litethinking](https://github.com/klipdigitalit/reto-uno-litethinking): https://github.com/klipdigitalit/reto-uno-litethinking

2. Configuración de GitHub Actions

.github/

workflows/

test-pipeline.yml

3. Configuración del Workflow: Se configuró el pipeline para ejecutarse automáticamente con cada Push a la rama principal.

4. Orquestación de componentes: Durante la ejecución del pipeline se automatizó el levantamiento de:

docker compose up -d – Kafka

npm run start-server – Backend

npm run start-worker – Worker de Kafka

vite --host 0.0.0.0 --port 5173 – Frontend

5. Ejecución de Playwright: npx playwright test tests/transfer-async.spec.js --workers=1 --repeat-each=3

6. Generación de evidencio

Se configuró la publicación automática del reporte Playwright como Artifact cuando ocurre una falla: if: failure()

Artifact generado: playwright-report.zip

Observación

Durante la implementación se identificaron varios problemas de infraestructura (permisos Linux, ejecución Headless, disponibilidad de servicios y sincronización Frontend/Backend) que fueron corregidos progresivamente hasta lograr una ejecución exitosa del pipeline, evidenciando una integración CI/CD funcional y alineada con las buenas prácticas de automatización E2E.